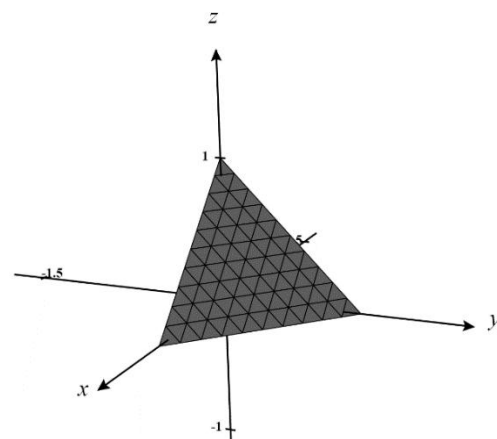


5.5 Surface Integrals Worksheet

1. Evaluate $\iint_{\sigma} xy \, dS$, where σ is the portion of the plane $x + y + z = 1$ in the first octant.

Since σ represents the graph of a function of the form $z = f(x, y)$, specifically $z = 1 - x - y$, we do not have to find a parameterization for σ and instead we tackle the problem as follows.

$$\begin{aligned} \iint_{\sigma} xy \, dS &= \iint_D \left[xy \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \right] dA \\ &= \iint_D \left[xy \sqrt{(-1)^2 + (-1)^2 + 1} \right] dA \\ &= \iint_D [xy\sqrt{3}] dA \end{aligned}$$

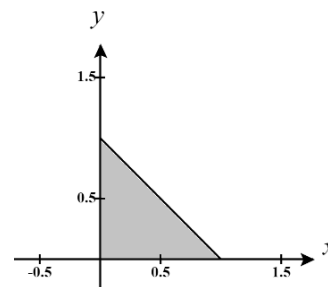


The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

$$D = \{(x, y) \mid 0 \leq y \leq 1 - x \text{ \& } 0 \leq x \leq 1\}$$

Therefore, we have that:

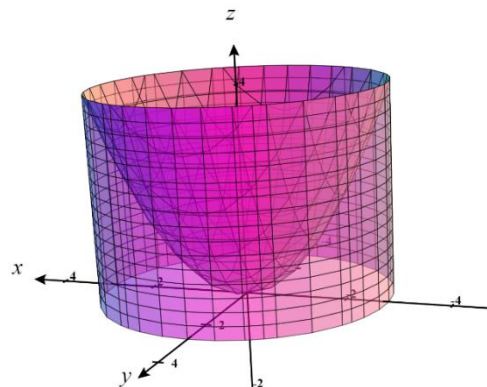
$$\begin{aligned} \iint_{\sigma} xy \, dS &= \iint_D [xy\sqrt{3}] dA = \int_0^1 \int_0^{1-x} xy\sqrt{3} \, dy \, dx \\ &= \int_0^1 \left. \frac{\sqrt{3}}{2} xy^2 \right|_0^{1-x} dx = \int_0^1 \frac{\sqrt{3}}{2} x(1-x)^2 dx = \frac{\sqrt{3}}{2} \int_0^1 (x - 2x^2 + x^3) dx \\ &= \frac{\sqrt{3}}{2} \left(\frac{1}{2}x^2 - \frac{2}{3}x^3 + \frac{1}{4}x^4 \right) \Big|_0^1 = \frac{\sqrt{3}}{2} \left(\frac{1}{2} - \frac{2}{3} + \frac{1}{4} \right) = \frac{\sqrt{3}}{2} \left(\frac{1}{12} \right) \\ &= \frac{\sqrt{3}}{24} \end{aligned}$$



2. Evaluate $\iint_{\sigma} \frac{1}{x^2 + y^2 + 1} dS$, where σ is the portion of the paraboloid $x^2 + y^2 = 2z$ inside the cylinder $x^2 + y^2 = 8$.

Since σ represents the graph of a function of the form $z = f(x, y)$, specifically $z = \frac{1}{2}(x^2 + y^2)$, we do not have to find a parameterization for σ and instead we tackle the problem as follows.

$$\begin{aligned} \iint_{\sigma} \frac{1}{x^2 + y^2 + 1} dS &= \iint_D \left[\frac{1}{1 + x^2 + y^2} \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} \right] dA \\ &= \iint_D \left[\frac{1}{1 + x^2 + y^2} \sqrt{x^2 + y^2 + 1} \right] dA \\ &= \iint_D \left[\frac{1}{\sqrt{1 + x^2 + y^2}} \right] dA \end{aligned}$$

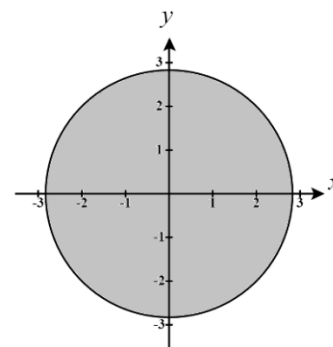


The region D is the projection of the solid into the xy -plane as shown in the picture to the right. We can view D in polar coordinates to make it easier to integrate over.

$$D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq \sqrt{8}\}$$

Therefore, we have that:

$$\iint_{\sigma} \frac{1}{x^2 + y^2 + 1} dS = \iint_D \left[\frac{1}{\sqrt{1 + x^2 + y^2}} \right] dA = \int_0^{2\pi} \int_0^{\sqrt{8}} \frac{1}{\sqrt{1 + r^2}} r dr d\theta$$



Let $u = 1 + r^2$ which implies $du = 2r dr \Rightarrow \frac{1}{2} du = r dr$. The bounds

become $u(0) = 1 + (0)^2 = 1$ and $u(\sqrt{8}) = 1 + (\sqrt{8})^2 = 9$.

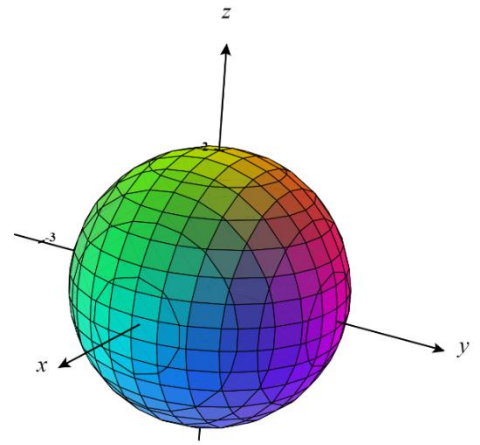
Therefore, we have

$$\begin{aligned} \int_0^{2\pi} \int_0^{\sqrt{8}} \frac{1}{\sqrt{1 + r^2}} r dr d\theta &= \int_0^{2\pi} \int_1^9 \frac{1}{2\sqrt{u}} du d\theta = \int_0^{2\pi} \int_1^9 \frac{1}{2} u^{-1/2} du d\theta \\ &= \int_0^{2\pi} u^{1/2} \Big|_1^9 d\theta = \int_0^{2\pi} 2 d\theta \\ &= 4\pi \end{aligned}$$

3. Evaluate $\iint_{\sigma} z^2 dS$ over the sphere $x^2 + y^2 + z^2 = 4$.

We see that we cannot write this surface as a function $z = f(x, y)$ and therefore we must approach this question by viewing the surface parametrically.

Parametric equations for σ are: $x = 2\sin(\phi)\cos(\theta)$,
 $y = 2\sin(\phi)\sin(\theta)$, $z = 2\cos(\phi)$ where $0 \leq \phi \leq \pi$ & $0 \leq \theta \leq 2\pi$ and
a vector equation which yields σ is
 $\mathbf{r}(\phi, \theta) = \langle \sin(\phi)\cos(\theta), \sin(\phi)\sin(\theta), \cos(\phi) \rangle$ where
 $0 \leq \phi \leq \pi$ & $0 \leq \theta \leq 2\pi$



In order to set up the surface integral as an iterated integral we need to know $\|\mathbf{r}_{\phi} \times \mathbf{r}_{\theta}\|$. So, we calculate that

$$\begin{aligned} \mathbf{r}_{\phi} \times \mathbf{r}_{\theta} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2\cos(\phi)\cos(\theta) & 2\cos(\phi)\sin(\theta) & -2\sin(\phi) \\ -2\sin(\phi)\sin(\theta) & 2\sin(\phi)\cos(\theta) & 0 \end{vmatrix} \\ &= \langle 4\sin^2(\phi)\cos(\theta), 4\sin^2(\phi)\sin(\theta), 4\cos(\phi)\sin(\phi)\cos^2(\theta) + 4\cos(\phi)\sin(\phi)\sin^2(\theta) \rangle \\ &= \langle 4\sin^2(\phi)\cos(\theta), 4\sin^2(\phi)\sin(\theta), 4\cos(\phi)\sin(\phi) \rangle. \end{aligned}$$

Then,

$$\begin{aligned} \|\mathbf{r}_{\phi} \times \mathbf{r}_{\theta}\| &= \sqrt{16\sin^4(\phi)\cos^2(\theta) + 16\sin^4(\phi)\sin^2(\theta) + 16\sin^2(\phi)\cos^2(\phi)} \\ &= 4\sqrt{\sin^4(\phi) + \sin^2(\phi)\cos^2(\phi)} = 4\sqrt{\sin^2(\phi)(\sin^2(\phi) + \cos^2(\phi))} = 4|\sin(\phi)|. \end{aligned}$$

Since we know that $0 \leq \phi \leq \pi$, we have that $|\sin(\phi)| = \sin(\phi)$.

Now, we have that

$$\iint_{\sigma} z^2 dS = \int_0^{2\pi} \int_0^{\pi} (2\cos(\phi))^2 4\sin(\phi) d\phi d\theta = \int_0^{2\pi} \int_0^{\pi} 16\cos^2(\phi)\sin(\phi) d\phi d\theta$$

let $u = \cos(\phi)$, then we note that $du = -\sin(\phi)d\phi \Rightarrow -du = \sin(\phi)d\phi$, $u(0) = 1$, and $u(\pi) = -1$. This means

$$\begin{aligned} &\int_0^{2\pi} \int_0^{\pi} 16\cos^2(\phi)\sin(\phi) d\phi d\theta \\ &= \int_0^{2\pi} \int_1^{-1} -16u^2 du d\theta = \int_0^{2\pi} \left[-\frac{16}{3}u^3 \right]_{u=1}^{u=-1} d\theta = \int_0^{2\pi} \frac{32}{3} d\theta \\ &= \frac{32}{3} \theta \Big|_0^{2\pi} = \frac{64\pi}{3} \end{aligned}$$

4. Consider the object which has the shape of the spiral ramp communicated as a parametric surface by $\mathbf{r}(u, v) = \langle v \cos(u), v \sin(u), u \rangle$, where $0 \leq u \leq 2\pi$ and $0 \leq v \leq 1$. Suppose that the density of the surface at a point is given by the function $f(x, y, z) = \frac{1}{1 + x^2 + y^2}$. Determine the mass of the object.

We calculate that:

$$\mathbf{r}_u = \langle -v \sin(u), v \cos(u), 1 \rangle$$

$$\mathbf{r}_v = \langle \cos(u), \sin(u), 0 \rangle$$

Therefore,

$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -v \sin(u) & v \cos(u) & 1 \\ \cos(u) & \sin(u) & 0 \end{vmatrix} \\ &= \langle -\sin(u), \cos(u), -v \sin^2(u) - v \cos^2(u) \rangle \\ &= \langle -\sin(u), \cos(u), -v \rangle \end{aligned}$$

and thus,

$$\|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{\sin^2(u) + \cos^2(u) + v^2} = \sqrt{1 + v^2}$$

Now, we have that,

$$\begin{aligned} \text{mass} &= \iint_{\sigma} f(x, y, z) dS = \iint_D f(x(u, v), y(u, v), z(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dA \\ &= \iint_D \frac{1}{1 + v^2 \cos^2(u)^2 + v^2 \sin^2(u)^2} \sqrt{1 + v^2} dA = \iint_D \frac{\sqrt{1 + v^2}}{1 + v^2} dA = \iint_D \frac{1}{\sqrt{1 + v^2}} dA \\ &= \int_0^{2\pi} \int_0^1 \frac{1}{\sqrt{1 + v^2}} dv du \end{aligned}$$

Evaluating this integral will require a trigonometric substitution.

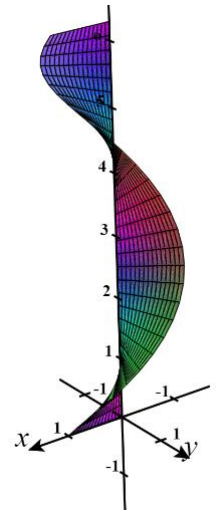
Let $v = \tan(\theta)$ and $dv = \sec^2(\theta) d\theta$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Our bounds become:

$$0 = \tan(\theta) \Rightarrow \theta = 0$$

$$1 = \tan(\theta) \Rightarrow \theta = \frac{\pi}{4}$$

We then get that

$$\begin{aligned} \int_0^1 \frac{1}{\sqrt{1 + v^2}} dv &= \int_0^{\pi/4} \frac{1}{\sqrt{1 + \tan^2(\theta)}} \sec^2(\theta) d\theta = \int_0^{\pi/4} \frac{1}{\sqrt{\sec^2(\theta)}} \sec^2(\theta) d\theta = \int_0^{\pi/4} \frac{1}{|\sec(\theta)|} \sec^2(\theta) d\theta \\ &= \int_0^{\pi/4} \frac{1}{\sec(\theta)} \sec^2(\theta) d\theta = \int_0^{\pi/4} \sec(\theta) d\theta \\ &= \ln |\sec(\theta) + \tan(\theta)| \Big|_0^{\pi/4} = \ln |\sqrt{2} + 1| - \ln |1 + 0| = \ln(\sqrt{2} + 1) \end{aligned}$$

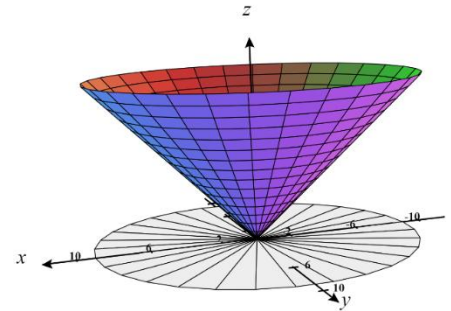


Therefore, we get that

$$\begin{aligned}
 mass &= \iint_D \frac{1}{\sqrt{1+v^2}} dA \\
 &= \int_0^{2\pi} \int_0^1 \frac{1}{\sqrt{1+v^2}} dv du \\
 &= \int_0^{2\pi} \ln(\sqrt{2}+1) du \\
 &= 2\pi \ln(\sqrt{2}+1)
 \end{aligned}$$

5. Let σ be the surface of the solid created by the cone $z = \sqrt{x^2 + y^2}$ below the plane $z = 9$.
- a) Use a surface integral to compute the surface area of σ *without* converting the surface to a parametric form.

$$\begin{aligned}
 SA &= \iint_{\sigma} 1 dS = \iint_D \sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 + 1} dA \\
 &= \iint_D \left[\sqrt{\left[\frac{1}{2}(x^2 + y^2)^{-1/2} 2x\right]^2 + \left[\frac{1}{2}(x^2 + y^2)^{-1/2} 2y\right]^2 + 1} \right] dA \\
 &= \iint_D \left[\sqrt{\frac{x^2}{x^2 + y^2} + \frac{y^2}{x^2 + y^2} + 1} \right] dA \\
 &= \iint_D \sqrt{2} dA
 \end{aligned}$$



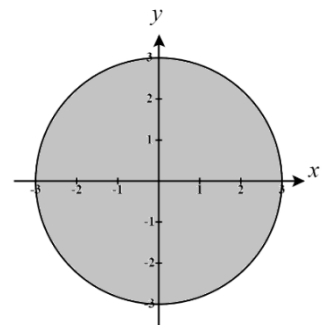
The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

We can view D in polar coordinates to make it easier to integrate over.

$$D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 9\}$$

Therefore, we have that:

$$\begin{aligned}
 SA &= \iint_{\sigma} 1 dS = \iint_D \sqrt{2} dA = \int_0^{2\pi} \int_0^9 \sqrt{2} r dr d\theta \\
 &= \int_0^{2\pi} \left. \frac{\sqrt{2}}{2} r^2 \right|_0^9 d\theta = \int_0^{2\pi} \frac{81\sqrt{2}}{2} d\theta \\
 &= 81\pi\sqrt{2}
 \end{aligned}$$



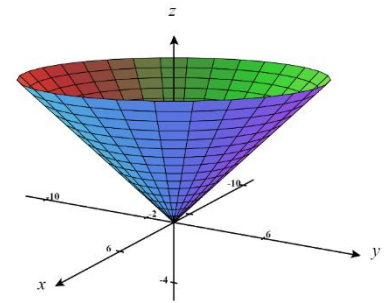
b) Parameterize the surface σ .

We see that as we progress up the z -axis we get circles of increasing radius. Therefore, we expect that z must vary freely while the values of x and y must create circles.

Therefore, we write create $\mathbf{r}(u, v) = \langle v \cos(u), v \sin(u), v \rangle$ where

$$0 \leq u \leq 2\pi \text{ \& } 0 \leq v \leq 9.$$

(We can check with the 3D plotter to see that our parametric surface is indeed the cone we have from the beginning of the question.)



c) Use a surface integral to compute the surface area σ using the parameterization you found in part (b).

We calculate that:

$$\mathbf{r}_u = \langle -v \sin(u), v \cos(u), 0 \rangle$$

$$\mathbf{r}_v = \langle \cos(u), \sin(u), 1 \rangle$$

Therefore,

$$\begin{aligned} \mathbf{r}_u \times \mathbf{r}_v &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -v \sin(u) & v \cos(u) & 0 \\ \cos(u) & \sin(u) & 1 \end{vmatrix} \\ &= \langle v \cos(u), v \sin(u), -v \sin^2(u) - v \cos^2(u) \rangle \\ &= \langle v \cos(u), v \sin(u), -v \rangle \end{aligned}$$

and thus,

$$\begin{aligned} \|\mathbf{r}_u \times \mathbf{r}_v\| &= \sqrt{v^2 \cos^2(u) + v^2 \sin^2(u) + v^2} \\ &= \sqrt{2v^2} \\ &= \sqrt{2}v \end{aligned}$$

Now, we have that,

$$\begin{aligned} SA &= \iint_{\sigma} 1 dS = \iint_D 1 \|\mathbf{r}_u \times \mathbf{r}_v\| dA = \int_0^9 \int_0^{2\pi} \sqrt{2}v du dv \\ &= \int_0^9 \sqrt{2}vu \Big|_0^{2\pi} dv \\ &= \int_0^9 2\pi\sqrt{2}v dv \\ &= \pi\sqrt{2}v^2 \Big|_0^9 = 81\pi\sqrt{2} \end{aligned}$$

6. Determine the surface area for the portion of the paraboloid $z = 4 - x^2 - y^2$ that sits above the xy -plane.

Since σ represents the graph of a function of the form $z = f(x, y)$ we do not have to find a parameterization for σ and instead we tackle the problem as follows.

$$\iint_{\sigma} 1 dS = \iint_D \left[\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2} + 1 \right] dA = \iint_D \left[\sqrt{4x^2 + 4y^2 + 1} \right] dA.$$

The region D is the projection of the solid into the xy -plane as shown in the picture to the right.

We can view D in polar coordinates to make it easier to integrate over.

$$D = \{(r, \theta) | 0 \leq \theta \leq 2\pi \text{ \& } 0 \leq r \leq 2\}$$

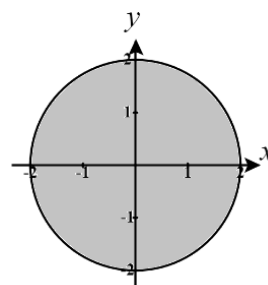
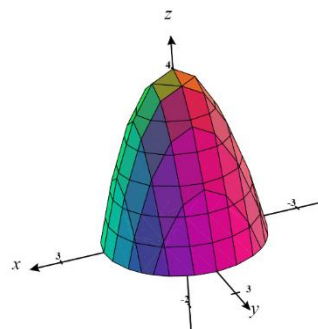
Therefore, we have that:

$$\iint_{\sigma} 1 dS = \iint_D \left[\sqrt{4x^2 + 4y^2 + 1} \right] dA = \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta$$

Let $u = 4r^2 + 1$ which implies $du = 8r dr \Rightarrow \frac{1}{8} du = r dr$. The bounds become

$u(0) = 4(0)^2 + 1 = 1$ and $u(2) = 4(2)^2 + 1 = 17$. Therefore, we have

$$\begin{aligned} \int_0^{2\pi} \int_0^2 \sqrt{4r^2 + 1} r dr d\theta &= \int_0^{2\pi} \int_1^{17} \frac{1}{8} \sqrt{u} du d\theta = \int_0^{2\pi} \frac{1}{12} u^{3/2} \Big|_1^{17} d\theta \\ &= \int_0^{2\pi} \frac{1}{12} (17^{3/2} - 1) d\theta \\ &= \frac{\pi}{6} (17^{3/2} - 1) \end{aligned}$$



7. Consider the cylindrical surface created by $x^2 + y^2 = 16$ between the planes $z = 1$ & $z = e$. Suppose that the density of the surface at a point is given by the function $f(x, y, z) = \ln(z)z + \frac{1}{16}(x^2 + y^2) - 1$. Determine the mass of the object.

We see that we cannot write this surface as a function $z = f(x, y)$ and therefore we must approach this question by viewing the surface parametrically.

We first parametrize the surface as $\mathbf{r}(u, v) = \langle 4 \cos(u), 4 \sin(u), v \rangle$ where $0 \leq u \leq 2\pi$ & $1 \leq v \leq e$.

We calculate that:

$$\mathbf{r}_u = \langle -4 \sin(u), 4 \cos(u), 0 \rangle$$

$$\mathbf{r}_v = \langle 0, 0, 1 \rangle$$

$$\text{Therefore, } \mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 \sin(u) & 4 \cos(u) & 0 \\ 0 & 0 & 1 \end{vmatrix} = \langle 4 \cos(u), 4 \sin(u), 0 \rangle$$

$$\text{and thus, } \|\mathbf{r}_u \times \mathbf{r}_v\| = \sqrt{16 \sin^2(u) + 16 \cos^2(u) + 0} = \sqrt{16} = 4$$

Now, we have that,

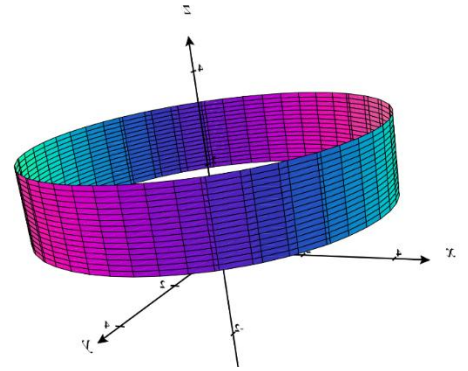
$$\begin{aligned} \text{mass} &= \iint_{\sigma} f(x, y, z) dS = \iint_D f(x(u, v), y(u, v), z(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dA \\ &= \iint_D \left[\ln(v)v + \frac{1}{16}(16 \cos^2(u) + 16 \sin^2(u)) - 1 \right] dA = \iint_D \ln(v)v dA \\ &= \int_0^{2\pi} \int_1^e \ln(v)v dv du \end{aligned}$$

Evaluating this integral will require integration by parts.

Let $k = \ln(v)$ and $dm = v dv$. This implies that $dk = \frac{1}{v} dv$ and $m = \frac{1}{2}v^2$.

Therefore, by IPB, we have that:

$$\begin{aligned} \int_1^e \ln(v)v dv du &= \frac{1}{2}v^2 \ln(v) \Big|_1^e - \int_1^e \frac{1}{2}v^2 \frac{1}{v} dv = \left(\frac{1}{2}e^2 - 0 \right) - \int_1^e \frac{1}{2}v dv \\ &= \frac{1}{2}e^2 - \left(\frac{1}{4}v^2 \right) \Big|_1^e = \frac{1}{2}e^2 - \left(\frac{1}{4}e^2 - \frac{1}{4} \right) \\ &= \frac{1}{4}e^2 + \frac{1}{4} \end{aligned}$$



Therefore, we get that

$$\begin{aligned} mass &= \iint_D \ln(v) v \, dA = \int_0^{2\pi} \int_1^e \ln(v) v \, dv \, du = \int_0^{2\pi} \left(\frac{1}{4} e^2 + \frac{1}{4} \right) du \\ &= \frac{\pi}{2} (e^2 + 1) \end{aligned}$$